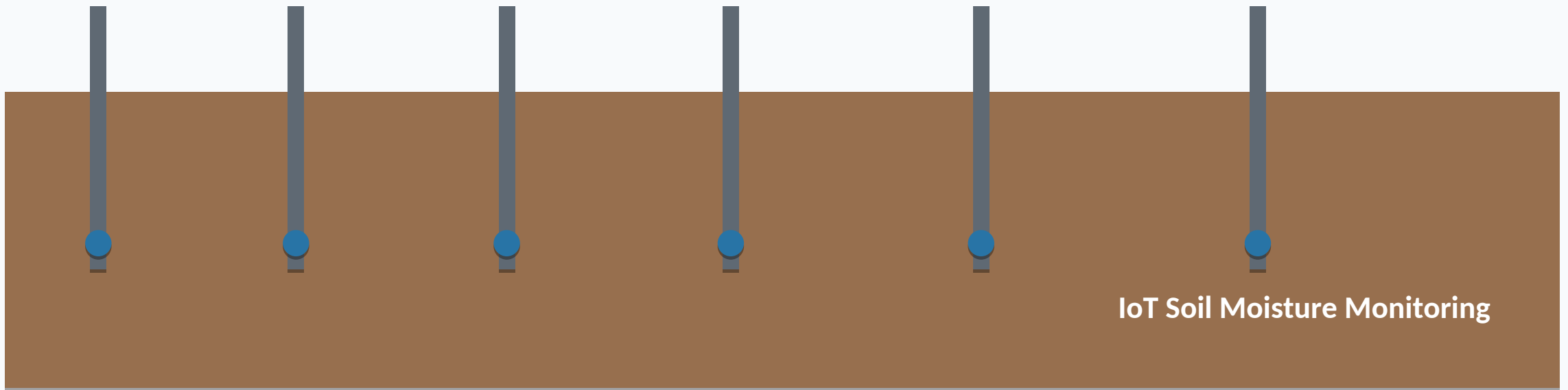


SOIL MOISTURE DATA

Flash Flood Prediction for Hilly Regions Using Multi-Source Data



1. What is Soil Moisture Data?

- Soil moisture is the amount of water currently stored in soil.
- It indicates whether the ground is dry, moderately wet, or close to saturation.
- Moisture changes with rainfall, evaporation, soil type, vegetation, and drainage.
- It is useful for estimating how much additional rainfall the soil can absorb.

DRY SOIL

More capacity to absorb incoming rain.

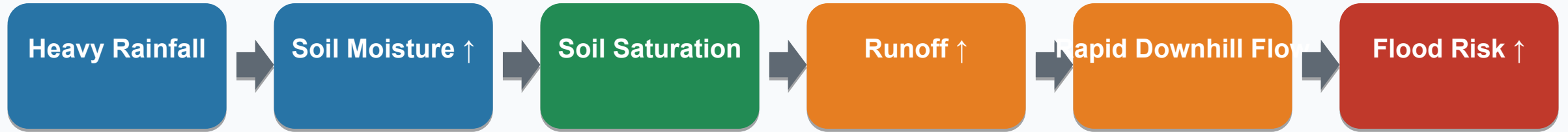
WET SOIL

Less remaining absorption capacity.

SATURATED SOIL

More rain can become surface runoff.

2. How Does Soil Moisture Contribute to Flash Floods?



Why it matters

When soil is already near saturation, additional rainfall may infiltrate less and produce more surface runoff. In hilly terrain, that runoff can move rapidly downslope.

3. Why Soil Moisture is Important in Hilly Regions

STEEP SLOPES

Water can travel quickly downhill, reducing the time available for response.

SATURATED GROUND

Already-wet soil has less capacity to absorb additional rainfall.

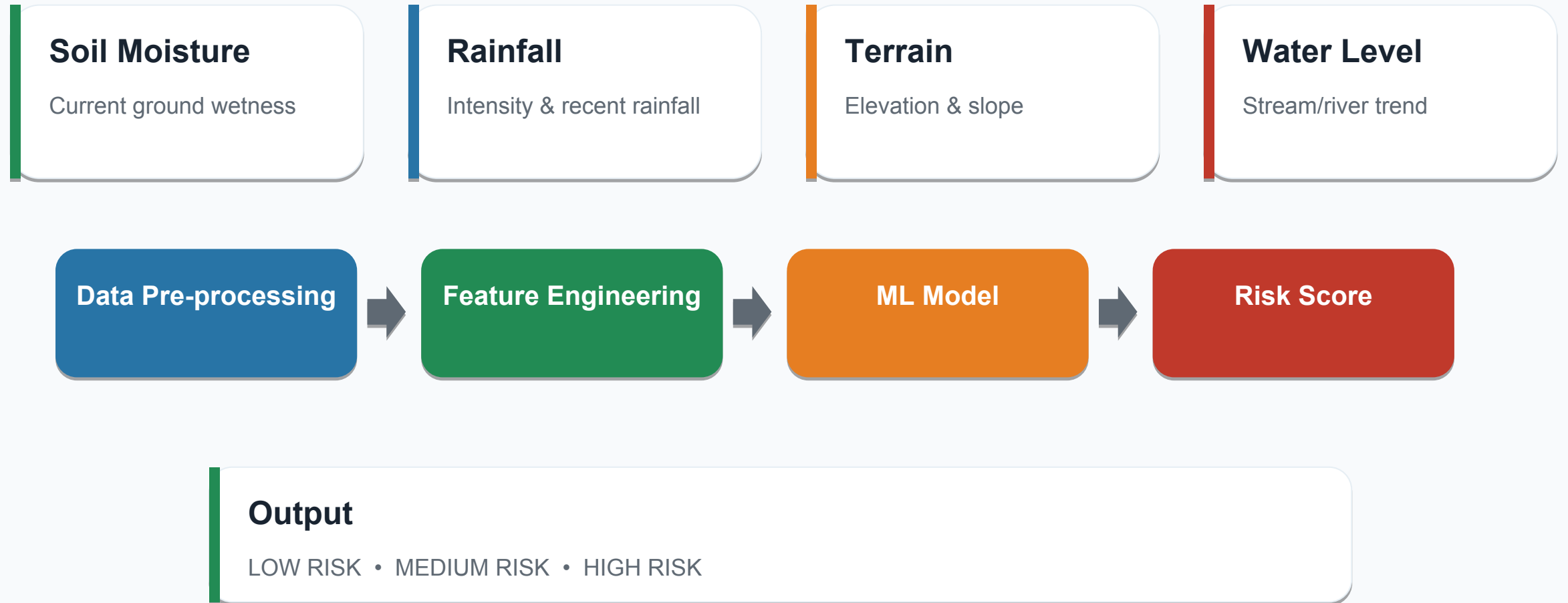
RAPID RUNOFF

Excess rainfall can collect in streams and channels quickly.

Key message

Soil moisture is not a standalone flood predictor. It becomes powerful when combined with rainfall, terrain, water level and weather data.

4. How Do We Predict Flash Flood Risk?



5. Where Can We Collect the Data?

Soil Moisture

IoT sensors; satellite/open Earth-observation datasets

Rainfall

India Meteorological Department (IMD) and suitable datasets

Elevation / Slope

Digital Elevation Models (DEM); slope can be derived from elevation

Water Level

Central Water Commission / State Water Resources sources where available

Historical Floods

NDMA / State Disaster Management records and suitable datasets

6. How is Soil Moisture Measured?

IoT / Field Sensor

A soil-moisture sensor is inserted into the ground. A microcontroller such as ESP32 reads the sensor and can send the value over Wi-Fi.

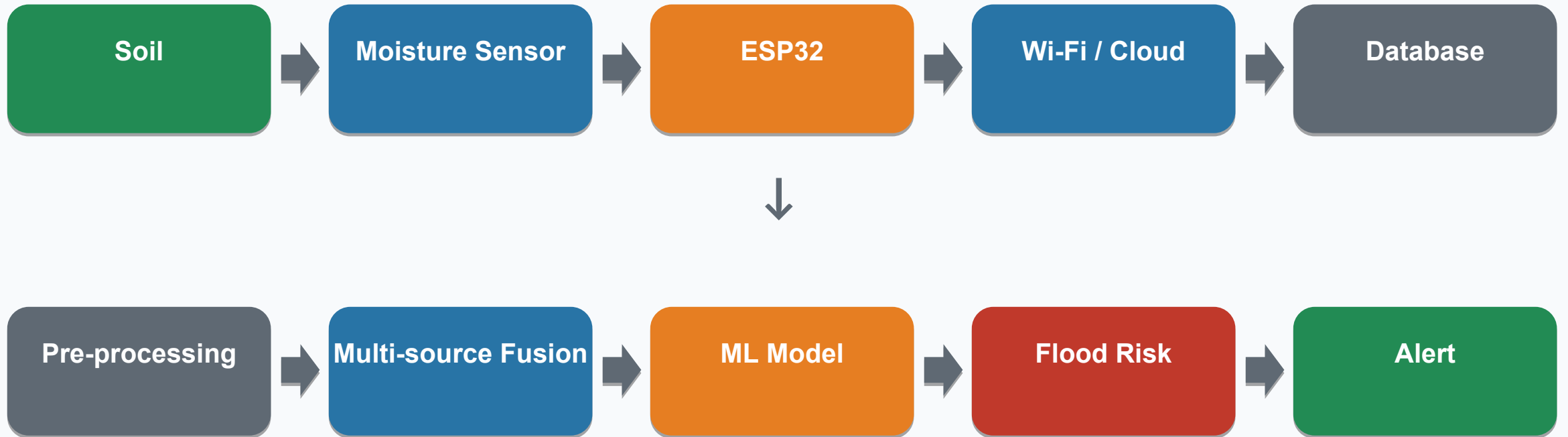
Satellite Data

Satellite products provide broader spatial coverage and can support historical or regional soil-moisture analysis.

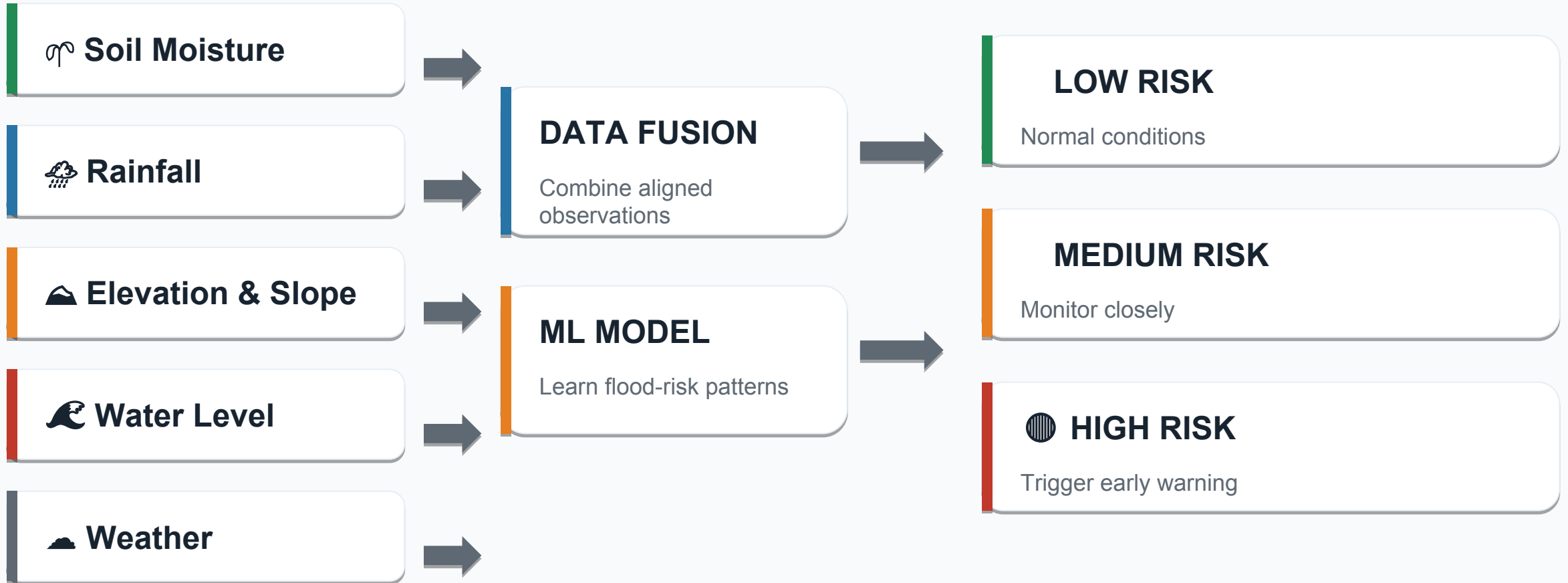
Recommended hackathon approach

Use IoT sensor readings for a live/local prototype and combine them with wider-area and historical data where available.

7. Soil Moisture IoT Architecture



8. Complete Multi-Source Prediction Architecture



9. Small IoT Prototype – ESP32 + Soil Moisture Sensor

```
#define SOIL_SENSOR 34

void setup() {
  Serial.begin(115200);
}

void loop() {
  int value = analogRead(SOIL_SENSOR);

  int moisture = map(value, 4095, 0, 0, 100);
  moisture = constrain(moisture, 0, 100);

  Serial.print("Soil Moisture: ");
  Serial.print(moisture);
  Serial.println("%");

  if (moisture < 30)
    Serial.println("Status: DRY");
  else if (moisture < 70)
    Serial.println("Status: MODERATE");
  else
    Serial.println("Status: HIGH / SATURATED");

  delay(2000);
}
```

Sensor Input

Analog reading from the soil-moisture sensor.

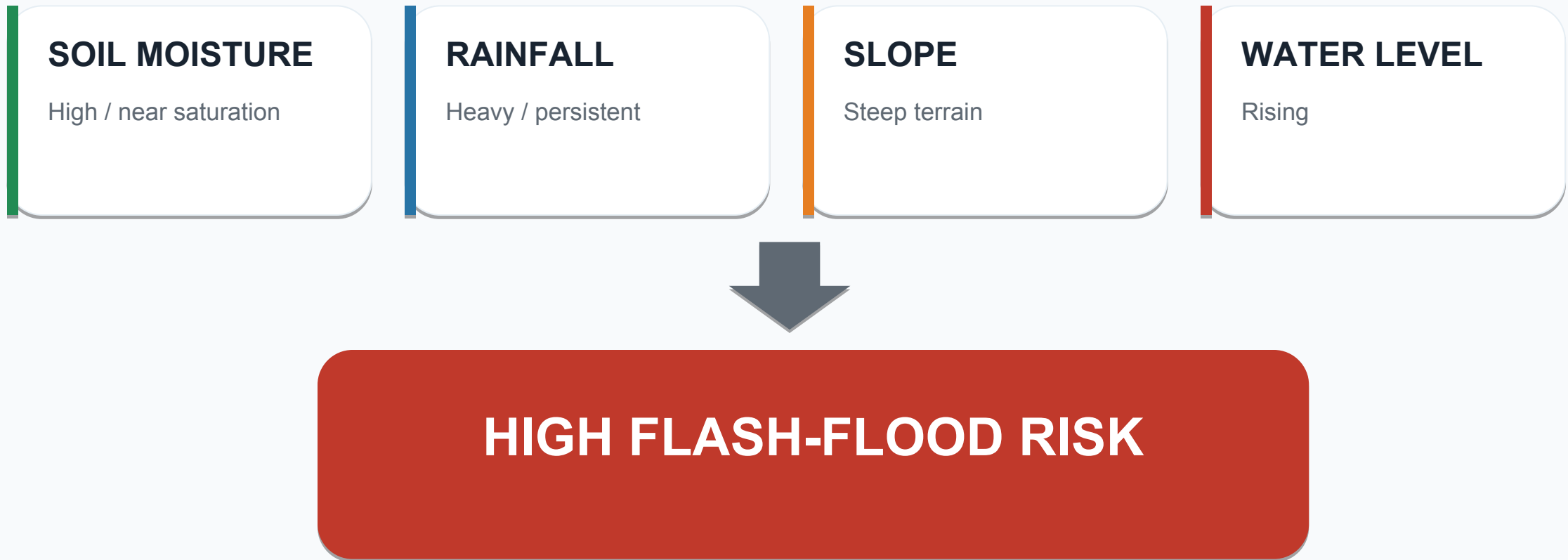
ESP32

Reads, converts and classifies the value.

Output

DRY / MODERATE / HIGH

10. Example of Flood-Risk Decision



This is an example decision pattern — the actual ML model must be trained and validated using historical data.

11. Conclusion & Future Scope

CONCLUSION

Soil moisture tells us how wet or saturated the ground is. Combining it with rainfall, terrain, water-level and weather data can improve flash-flood risk assessment in hilly regions.

FUTURE SCOPE

- Real-time IoT sensor network
- GIS-based risk maps
- Better ML / deep-learning models
- Mobile and SMS alerts
- Integration with disaster-management systems

Thank You

Questions & Discussion